ALICE L WALTON SCHOOL OF MEDICINE

Diseases of the Immune System: Transplant Rejection

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Adapted from: Transplantation

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Learning Objectives (2) Versions

After completing this brick, you will be able to:

1. Compare and contrast the types of rejection (hyperacute, acute, and chronic) by clinical and histopathologic findings.
2. Describe graft-versus-host disease.

CASE CONNECTION

RP is a 60-year-old man with acute myeloid leukemia who is being considered for a bone marrow transplant. As you discuss this possibility with him, he says, "I've heard of a kidney transplant and a heart transplant but never a bone marrow transplant," RP says. "It sounds interesting, but what are the risks?"

How will you respond to RP's concern? Consider your answer as you read, and we'll revisit at the end of the brick.

GO TO CONCLUSION ↓

What Are the Three Types of Transplant Rejection (LO #1)?

If the recipient's immune system sees the graft as foreign, the recipient's T and B cells will mount an immune response in an attempt to reject (destroy) the graft. There are three types of graft rejection based on clinical and pathologic features: hyperacute, acute, and chronic.

Hyperacute rejection occurs within minutes and up to 24 hours after transplant. This type of rejection occurs when the recipient already has **pre-formed antibodies** that react against ABO blood group antigens or type I MHC molecules present on the graft endothelial cells (Figure 1). These antibodies trigger the complement cascade, leading to ischemia and necrosis of the transplanted organ or tissue. This is known as a type 2 hypersensitivity reaction. The graft must be removed as soon as possible.

CLINICAL CORRELATION

The antibodies may be natural IgM antibodies against a blood group antigen or antibodies specific for MHC molecules that were induced by prior exposure (blood transfusion, pregnancy, transplantation of another organ).

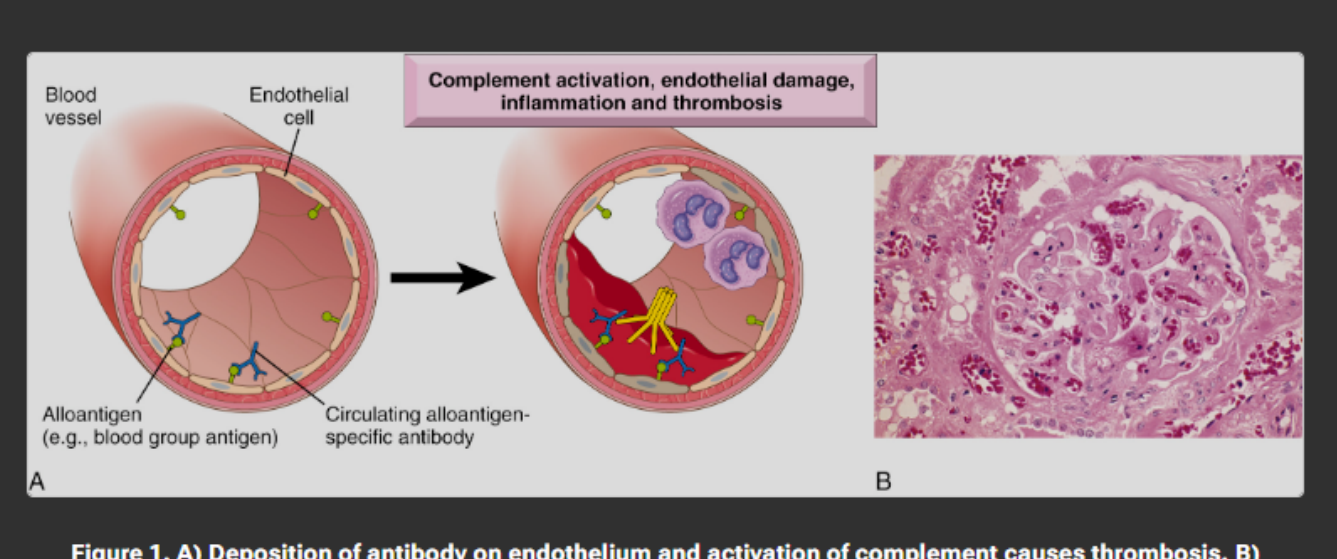
Blood vessel
Endothelial cell
Complement activation, endothelial damage, inflammation and thrombosis
Alloantigen (e.g., blood group antigen)
Circulating alloantigen-specific antibody
A
B

Figure 1. A) Deposition of antibody on endothelium and activation of complement causes thrombosis. B) Hyperacute rejection of a kidney allograft showing platelet fibrin thrombi and severe ischemic injury in a glomerulus.

ACTIVE LEARNING QUESTION

Review: Match the following hypersensitivity reaction with the mechanism.

Type I Hypersensitivity	✓ Antibody-mediated; production of IgE antibody causing release of mast cell mediators
Type II Hypersensitivity	✓ Antibody-mediated; production of IgG or IgM antibodies injure cells by promoting their phagocytosis or lysis and induce inflammation
Type III Hypersensitivity	✓ Antibody-mediated; Deposition of immune complexes (IgG or IgM antibodies) in tissues induces inflammation and tissue damage
Type IV Hypersensitivity	✓ Cell-mediated; T cells cause direct cytotoxicity and release of cytokines, inflammation, and macrophage activation

Explanation

Type I hypersensitivity reactions are immediate, with injury being caused by IgE antibodies that cause the release of mast cell mediators. These mediators act on vessels and smooth muscle cells (vasodilation, edema, smooth muscle contraction, mucus production). This reaction is seen with anaphylaxis, allergies, and bronchial asthma. Type II hypersensitivity reactions occur when IgG and IgM antibodies injure cells by promoting their phagocytosis and induce inflammation. Examples of this include autoimmune hemolytic anemia, Goodpasture s ion. Type III hypersensitivity reactions occur when antigen-antibody complexes are deposited in tissues and induce inflammation and tissue damage. Examples of this include lupus and post-streptococcal glomerulonephritis. Type IV hypersensitivity reactions occur when T-cells cause either direct cytotoxicity (CD8+ T cells) or induce inflammation/cytokine release (CD4+ T cells). Examples of this include contact dermatitis, type I diabetes, and graft-versus-host disease.

Acute rejection occurs 1 week to several months after transplant and occurs via both cell-mediated (T-cell) and antibody-mediated (B-cell) mechanisms. It is the principal cause of early graft failure and responds to immunosuppressive medication. Acute rejection is divided into two types, though most patterns may be present in a given patient.

- **Acute cellular rejection:** CD8+ T cells directly destroy graft cells or CD4+ T cells secrete cytokines and induce inflammation, damaging the graft (Figure 2).

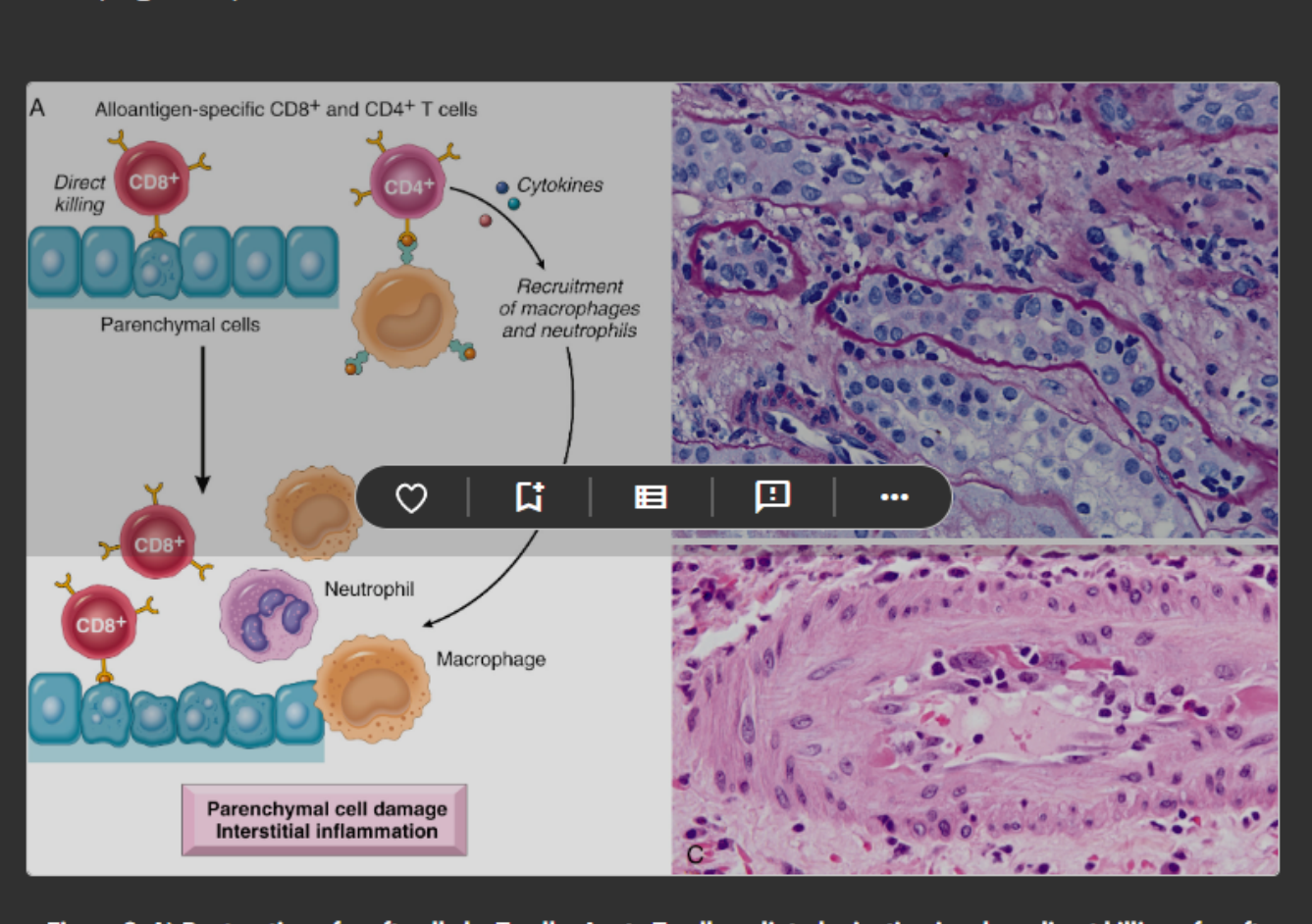
A
Alloantigen-specific CD8+ and CD4+ T cells
Direct killing
CD8+
CD4+
Cytokines
Parenchymal cells
Recruitment of macrophages and neutrophils
Neutrophil
Macrophage
Parenchymal cell damage Interstitial inflammation
C

Figure 2. A) Destruction of graft cells by T cells. Acute T-cell mediated rejection involves direct killing of graft cells by CD8+ T-cells and inflammation caused by cytokines produced by CD4+ T cells. B) Acute cellular rejection of a kidney graft, manifested by inflammatory cells in the interstitium and between epithelial cells of the tubules (tubilitis). Collapsed tubules are outline by wavy basement membranes. C) Rejection vasculitis in a kidney graft. An arteriole is shown with inflammatory cells attacking and undermining the endothelium (endothelitis).

- **Acute antibody-mediated rejection:** antibodies bind to vascular endothelium and activate complement; the resultant inflammation and endothelial damage cause graft failure (Figure 3).

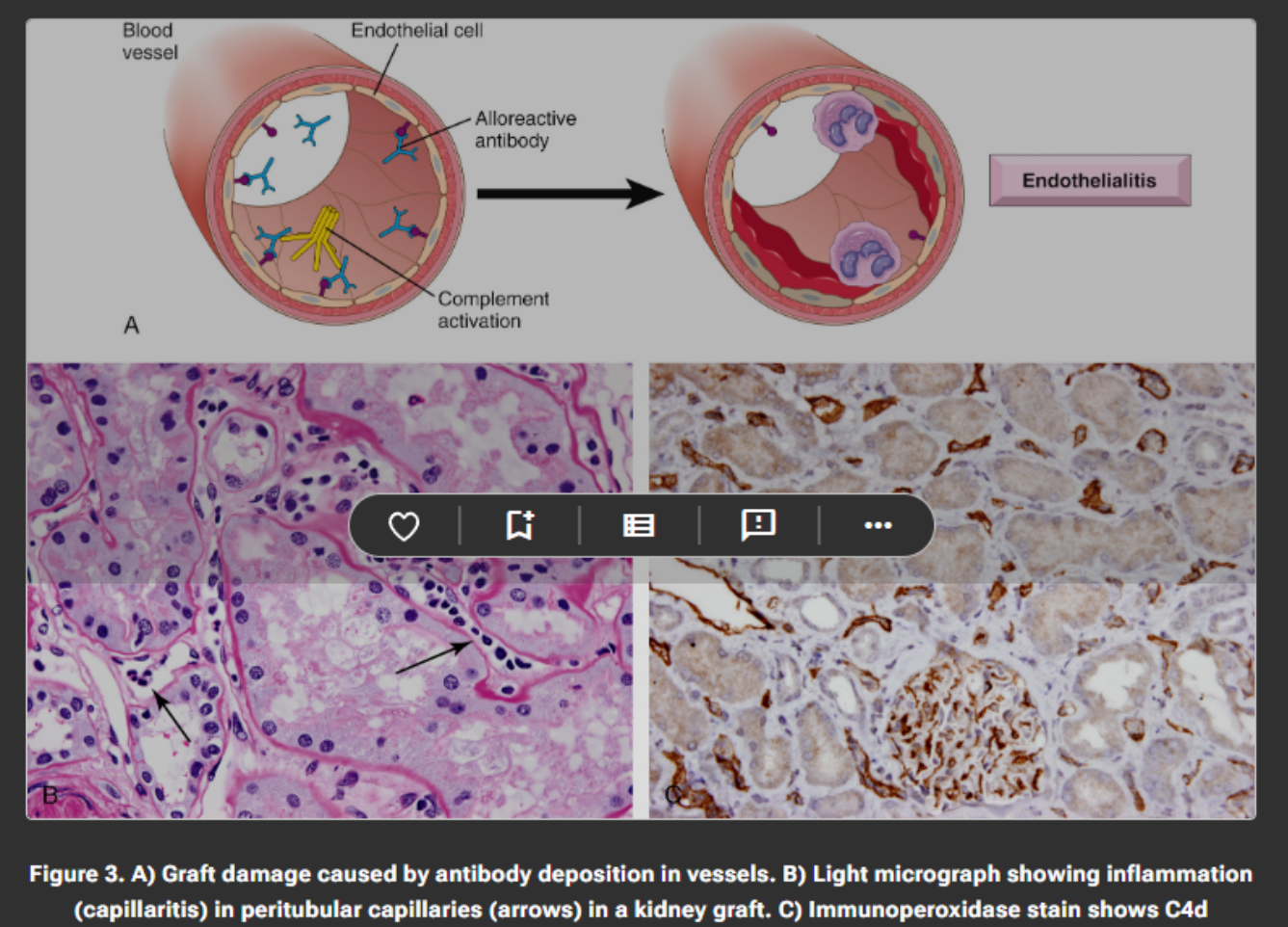

Figure 3. A) Graft damage caused by antibody deposition in vessels. B) Light micrograph showing inflammation (capillaritis) in peritubular capillaries (arrows) in a kidney graft. C) Immunoperoxidase stain shows C4d deposition in peritubular capillaries and a glomerulus.

CLINICAL CORRELATION

It is common for transplant recipients to be taking four or more immunosuppressive drugs in the days immediately after surgery. Most recipients require two to three drugs throughout their lifespan to prevent rejection. The drugs suppress the immune system, so these patients are at risk for infections.

Chronic rejection can occur months to years after transplant. Like acute rejection, this process involves both cellular and humoral immune responses (Figure 4). Immunosuppressive drugs act to prevent chronic rejection.

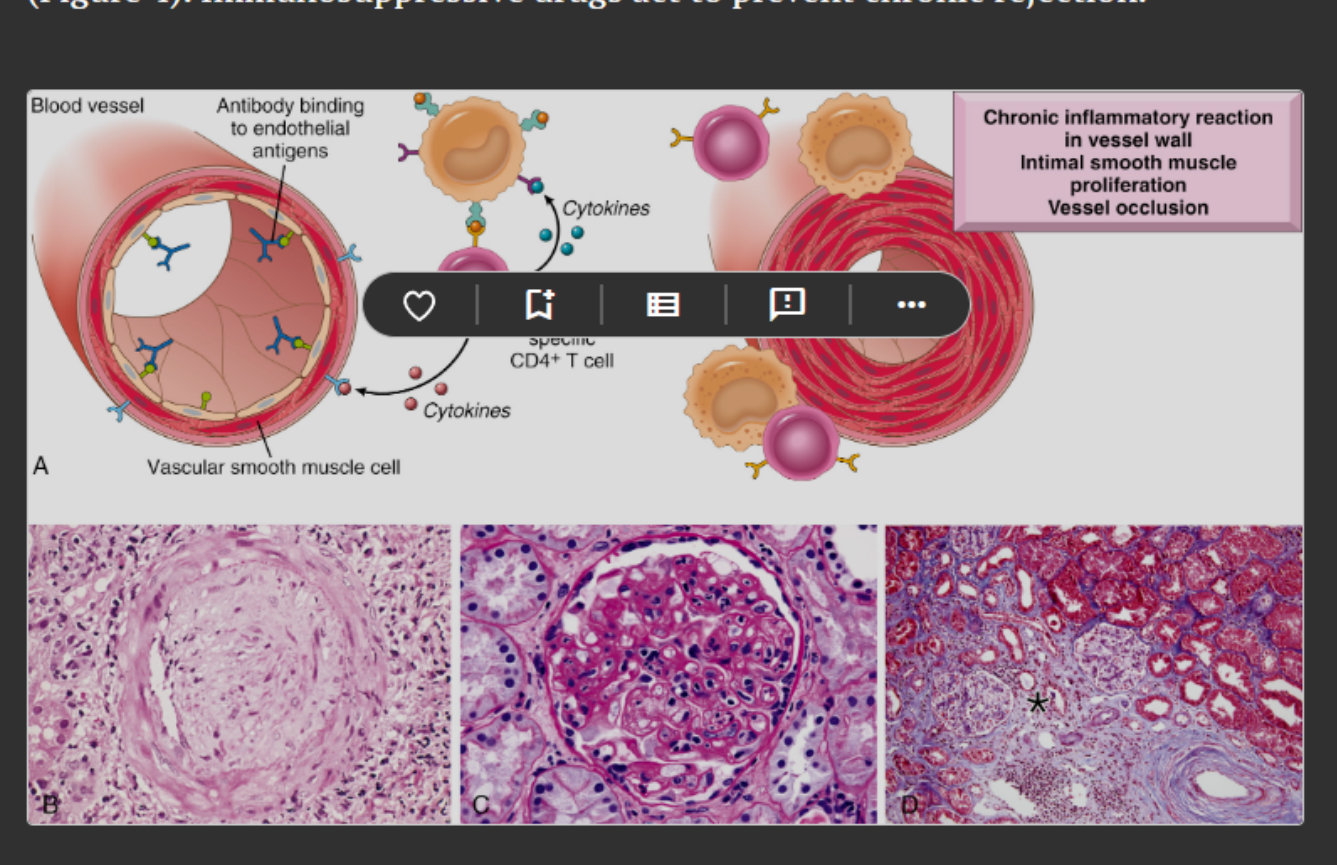
Blood vessel
Antibody binding to endothelial antigens
Cytokines
CD4+ T cell
Cytokines
Vascular smooth muscle cell
Chronic inflammatory reaction in vessel wall
Intimal smooth muscle proliferation
Vessel occlusion
A
B
C
D

Figure 4. A) Graft arteriosclerosis caused by T-cell cytokines and antibody deposition. B) Graft arteriosclerosis in a cardiac transplant. C) The glomerulus shows inflammatory cells within the capillary loops (glomerulitis), accumulation of mesangial matrix, and duplication of the capillary basement membrane. D) Interstitial fibrosis and tubular atrophy. In this trichrome stain, the blue area (asterisk) shows fibrosis, contrasted with the normal kidney (top right).

CLINICAL CORRELATION

Several histologic criteria have been created to diagnose and grade rejection for organ transplants (ISHLT for heart transplant, Banff criteria for kidney transplants).

Which types of transplant rejection are mediated by antibodies?

Graft-Versus-Host Disease (LO #2)

Graft-versus-host disease (GVHD) is a potentially fatal complication that typically occurs in transplantation of allogeneic bone marrow (but is rare in transplantation of other types of organs). In an allogeneic bone marrow transplant, bone marrow cells from a non-identical donor are transplanted into a recipient whose bone marrow has been eradicated with chemotherapy to remove cancerous cells and provide a "clean slate" for the transplant. Unlike the graft rejection process, in which the *recipient's* immune cells recognize the *graft* tissue as foreign, in GVHD, immune cells from the *graft* recognize the *recipient's* tissue as foreign.

Bone marrow transplant tissue typically contains a lot of T cells, which—as discussed above—are immune system cells that identify and destroy foreign invaders such as viruses and bacteria. In GVHD, the donor's transplanted T cells attack the recipient's organs and tissues, resulting in severe organ dysfunction.

GVHD occurs most frequently in recipients who have been exposed to radiation or who have leukemia, autoimmune diseases, or immunodeficiencies. Other contributing factors i lantation from female donors to male recipients.

CLINICAL CORRELATION

Some of these risk factors may be due to immunosuppression of the recipient, leading to a decrease in the ability of the recipient to neutralize the T-lymphocyte attack from the donor.

Patients with GVHD may present with a maculopapular rash (may lead to desquamation in severe cases), jaundice (yellowing of the skin and mucous membranes; caused by destruction of small bile ducts), nausea, diarrhea (caused by mucosal ulceration of the gut), vomiting, abdominal pain, and hepatosplenomegaly (enlarged liver and spleen) (Figure 5). Symptoms may appear within weeks after transplant (especially the skin, liver, and gastrointestinal tract) or later, up to several years after transplant. Immunosuppressive drugs are used to both prevent and treat GVHD.

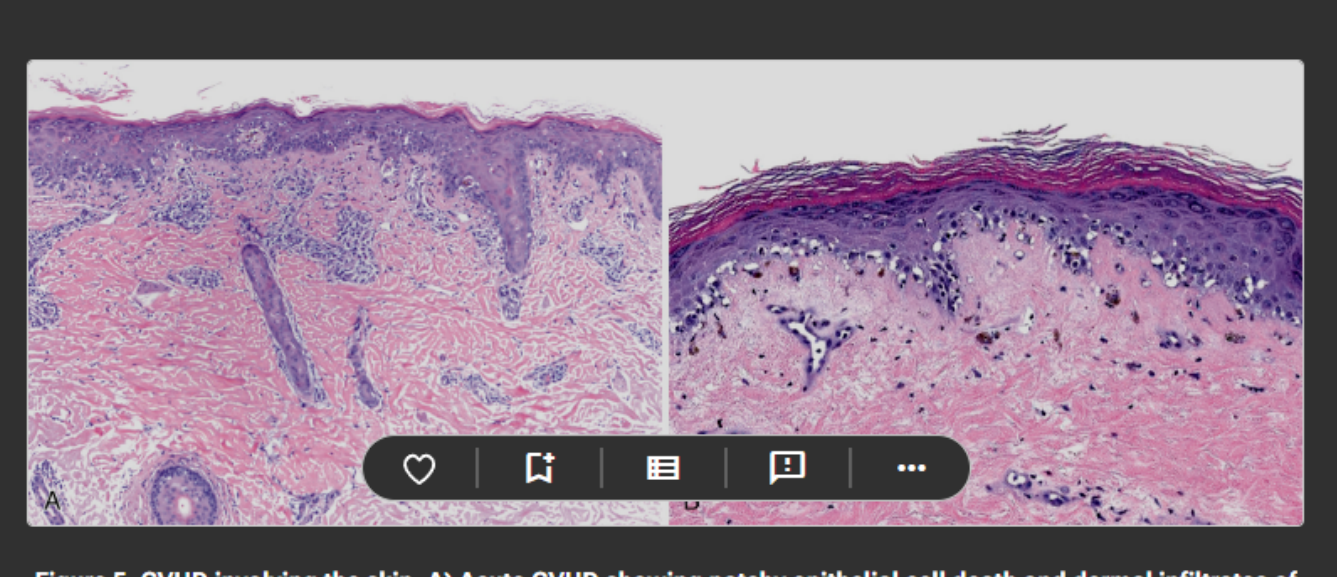

Figure 5. GVHD involving the skin. A) Acute GVHD showing patchy epithelial cell death and dermal infiltrates of lymphocytes and macrophages. B) Chronic GVHD showing a sparse lymphocytic infiltrate at the dermal-epidermal junction. The epidermis is thinned, signifying atrophy. The underlying dermis shows thickening of collagen bundles, indicative of sclerosis.

CLINICAL CORRELATION

Although tissue injury can be severe, affected tissues are not usually heavily infiltrated by lymphocytes. It is believed that damage is caused by cytokine release in addition to direct cytotoxicity by CD8+ T cells.

Which cells cause graft-versus-host disease?

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to RP, how do you respond to his concern about the risks of bone marrow transplant?

You explain the risks of infection because of the chemotherapy and radiation therapy RP will be given before the transplant. "It's important to wipe out the cancer cells from your bone marrow before we give you the healthy bone marrow of the donor. Another complication is graft-versus-host disease. This can occur when the cells of the donor attack the cells of the recipient because they recognize them as foreign. But giving you medications to suppress your immune system will help mitigate this risk."

Summary

- Transplant rejection is classified into three types: hyperacute, acute, and chronic, each with distinct clinical and pathologic features.
- Hyperacute rejection occurs within minutes to 24 hours post-transplant due to pre-formed antibodies in the recipient that react against the graft, leading to ischemia and necrosis.
- Acute rejection occurs 1 week to several months post-transplant and is caused by both cell-mediated (T-cell) and antibody-mediated (B-cell) mechanisms.
- Chronic rejection can occur months to years post-transplant and involves both cellular and humoral immune responses.
- Graft-versus-host disease (GVHD) is a potentially fatal complication of allogeneic bone nor's T cells attack the recipient's organs and tissues.
- Immunosuppressive drugs are used post-transplant to prevent rejection and treat GVHD, but they also increase the risk of infections.

Review Questions

1. A 52-year-old man received an allogeneic bone marrow transplant for acute leukemia. Two weeks after the transplant, the patient noticed a new skin rash along with diarrhea. What is most likely causing his symptoms?

A. ○ Donor B cells
B. ● Donor T cells
C. ○ Recipient antibodies
D. ○ Recipient antigens
E. ○ Recipient T cells

Hide Explanation

The correct answer is donor T cells (B). This is a two-step question. First, you must recognize that these symptoms are due to graft-versus-host disease after the bone marrow transplant. Then you must remember that this causes a type 4 hypersensitivity reaction, which is mediated by donor T cells. GVHD has symptoms caused by the donor T cells. Donor B cells (A) have not been found to cause these symptoms in GVHD. T cells (E) do not cause the symptoms of GVHD.

2. A 45-year-old woman received a kidney transplant to manage chronic renal failure due to diabetic nephropathy. After the kidney was placed, the surgeon noticed intraoperatively that the organ was pale and not receiving the proper blood supply. An inflammatory reaction was noted as the cause of a blood clot that had formed, preventing perfusion of the transplanted organ. Which of the following is the likely cause of this complication?

A. ○ Acute rejection
B. ○ Chronic rejection
C. ○ Graft-versus-host disease
D. ● Hyperacute rejection
E. ○ Subacute rejection

Hide Explanation

The correct answer is hyperacute rejection (D). The type of rejection described is hyperacute because it occurred within minutes of transplantation. Acute rejection (A) occurs 1-2 weeks after the transplant, while chronic rejection (B) occurs months to years later. Graft-versus-host disease (C) occurs weeks to months after transplant and is most likely to result from a bone marrow transplant, not a kidney transplant. Subacute rejection (E

How would you rate this Brick?

References

- Kumar V, Abbas AK, Aster JC, eds. *Robbins and Cotran Pathologic Basis of Disease.* 10th ed. Elsevier; 2021.

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Compare and contrast the types of rejection (hyperacute, acute, and chronic) by clinical and histopathologic findings.~~
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Objective Confidence: 2/2

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